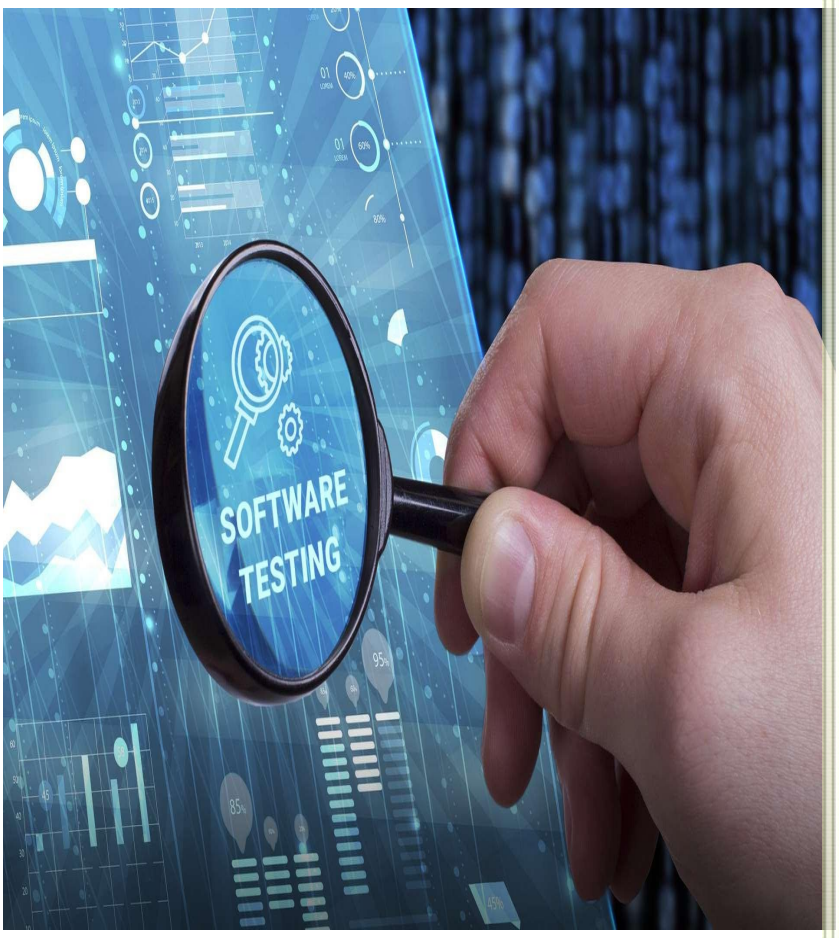
Software Testing Lab Manual



P.E.S.College of Engineering, Mandya-571401

Department of Information Science & Engineering

SOFTWARE TESTING LAB MANUAL /P21ISL606

Vision

"The department strives to equip our graduates with Knowledge and Skills to contribute significantly to Information Science & Engineering and enhance quality research for the benefit of society".

Mission

M1: To provide students with state of art facilities and tools of Information Science & Engineering to become productive, global citizens and life-long learners.

M2: To prepare students for careers in IT industry, Higher education and Research.

M3: To inculcate leadership qualities among students to make them competent Information Science & Engineering professionals or entrepreneurs.

Program Educational Objectives (PEOs)

Graduates of the program will be able to

PEO1: Establish a productive Information Science & Engineering career in industry, government or academia.

PEO2: Interact with their peers in other disciplines by exhibiting professionalism and team work to contribute to the economic growth of the country.

PEO3: Promote the development of solutions to the problems in Information Science using hardware and software integration.

PEO4: Pursue higher studies in Engineering, Management or Research.

Program Outcomes (POs)

PO1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems

PO2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Prerequisite

Instruction: 1. Basics of C coding Knowledge

2. Eclipse IDE and java programming knowledge for running test scenarios.

Course Outcomes (COs) :
1. Apply the appropriate test cases for any given problem.
2. Implement Test Automation process and experiment with testing tools.

List of Experiments

Sl.No	Experiments Title
	Part-A
1	Design, develop, code and run the program in any suitable language to solve the commission problem. Analyze it from the perspective of boundary value testing, derive different test cases, execute these test cases and discuss the test results.
2	Design, develop, code and run the program in any suitable language to implement the Next Date function. Analyze it from the perspective of boundary value testing, derive different test cases, execute these test cases and discuss the test results.
3	Design, develop, code and run the program in any suitable language to solve the commission problem. Analyze it from the perspective of equivalence class testing, derive different test cases, execute these test cases and discuss the test results.
4	Design, develop, code and run the program in any suitable language to implement the NextDate function. Analyze it from the perspective of equivalence class value testing, derive different test cases, execute these test cases and discuss the test results.
5	Design , Develop ,Code & Run the program in any suitable language to implement triangle problem
	Part-B
6	Write a script to login a specific web page.
7	Write a script to count number of items present on a desktop.
8	Write a script to update 06 student records into table into Excel file.
9	Write a script to provide total number of objects present / available on the page.
10.	Write a script to get the number of items in a list / combo box.

References:

1. Selenium Tutorial for Beginners: An Introduction to Selenium (stackify.com)

Course Articulation Matrix

Co/PO	PO1	PO2	PO31	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO 1	2	2	2	2								
CO 2	2	2	2	2	2							

Lab Program: 1

Design, develop, code and run the program in any suitable language to solve the commission problem. Analyze it from the perspective of boundary value testing, derive different test cases, execute these test cases and discuss the test results.

ALGORITHM

STEP 1: Define lockPrice=45.0, stockPrice=30.0, barrelPrice=25.0

STEP2: Input locks

STEP3: while(locks!=-1) ‘input device uses -1 to indicate end of data

goto STEP 12

STEP4:input (stocks, barrels)

STEP5: compute lockSales, stockSales, barrelSales and sales

STEP6: output(“Total sales:” sales)

STEP7: if (sales > 1800.0) goto STEP 8 else goto STEP 9

STEP8: commission=0.10*1000.0; commission=commission+0.15 * 800.0;

commission = commission + 0.20 * (sales-1800.0)

STEP9: if (sales > 1000.0) goto STEP 10 else goto STEP 11

STEP10: commission=0.10* 1000.0; commission=commission + 0.15 *

(sales-1000.0)

STEP11: Output(“Commission is $”, commission)

STEP12: exit

PROGRAM CODE:

```
#include<stdio.h>

#include<conio.h>

int main()

{

int c1,c2,c3,temp;

int locks,stocks,barrels,totallocks,totalstocks,totalbarrels;

float lockprice,stockprice,barrelprice,locksales,stocksales,barrelsales,sales,com;
```

```
lockprice=45.0;
stockprice=30.0;
barrelprice=25.0;
totallocks=0;
totalstocks=0;
totalbarrels=0;
clrscr();
printf("Enter the number of locks and to exit press -1\n");
scanf("%d",&locks);
while(locks != -1)
{
c1=(locks<=0 || locks>70);
printf("\nEnter the number of stocks and barrels\n");
scanf("%d %d",&stocks,&barrels);
c2=(stocks<=0 || stocks>80);
c3=(barrels<=0 || barrels>90);
if(c1)
printf("\nValue of locks are not in the range of 1.... 70\n");
else
{
temp=totallocks+locks;
if(temp>70)
printf("New totallocks = %d not in the range of 1.....70\n",temp);
else
totallocks=temp;
}
printf("Total locks = %d",totallocks);
if(c2)
```

```
printf("\n Value of stocks not in the range of 1.... 80\n");
else
{
temp=totalstocks+stocks;
if(temp>80)
printf("\nNew total stocks = %d not in the range of 1.... 80",temp);
else
totalstocks=temp;
}
printf("\nTotal stocks = %d",totalstocks);
if(c3)
printf("\n Value of barrels not in the range of 1.....90\n");
else
{
temp=totalbarrels+barrels;
if(temp>90)
printf("\nNew total barrels = %d not in the range of 1.....90\n",temp);
else
totalbarrels=temp;
}
printf("\nTotal  barrels=%d",  totalbarrels);
printf("\nEnter the number of locks and to exit press -1\n");
scanf("%d",&locks);
}
printf("\n Total locks = %d",totallocks);
prin tf("\n Total stocks = %d",totalstocks);
printf("\n Total barrels = %d",totalbarrels);
locksales=totallocks*lockprice;
```

```
stocksales=totalstocks*stockprice;
barrelsales=totalbarrels*barrelprice;
sales=locksales+stocksales+barrelsales;
printf("\n Total sales = %f",sales);
if(sales>1800)
{ com=0.10*100
0;
com=com+(0.15*800);
com=com+0.20*(sales-1800);
}
else if(sales>1000)
{ com=0.10*100
0;
com=com+0.15*(sales-1000);
}
else
com=0.10*sales;
printf("\nCommission = %f",com);
getch();
return 0;
}
```

Lab Program: 2

Design, develop, code and run the program in any suitable language to implement the Next Date function. Analyze it from the perspective of boundary value testing, derive different test cases, execute these test cases and discuss the test results.

ALGORITHM

STEP 1: Input date in format DD.MM.YYYY

STEP2: if MM is 01, 03, 05,07,08,10 do STEP3 else STEP6

STEP3:if DD < 31 then do STEP4 else if DD=31 do STEP5 else

output(Invalid Date);

STEP4: tomorrowday=DD+1 goto STEP18

STEP5: tomorrowday=1; tomorrowmonth=month + 1 goto STEP18

STEP6: if MM is 04, 06, 09, 11 do STEP7

STEP7: if DD<30 then do STEP4 else if DD=30 do STEP5 else

output(Invalid Date);

STEP8: if MM is 12

STEP9: if DD<31 then STEP4 else STEP10

STEP10: tomorrowday=1, tommorowmonth=1,

tommorowyear=YYYY+1; goto STEP18

STEP11: if MM is 2

STEP12: if DD<28 do STEP4 else do STEP13

STEP13: if DD=28 & YYYY is a leap do STEP14 else STEP15

STEP14: tommorowday=29 goto STEP18

STEP15: tommorowday=1, tomorrowmonth=3, goto STEP18;

STEP16: if DD=29 then do STEP15 else STEP17

STEP17: output("Cannot have feb", DD); STEP19

STEP18: output(tomorrowday, tomorrowmonth, tomorrowyear);

STEP19: exit

PROGRAM CODE:

```
#include<stdio.h>
```

```
#include<conio.h>
main( )
{
int
month[12]={31,28,31,30,31,30,31,31,30,31,30,31};
int d,m,y,nd,nm,ny,ndays;
clrscr( );
printf("enter the date,month,year");
scanf("%d%d%d",&d,&m,&y);
ndays=month[m-1];
if(y<=1812 && y>2012)
{
printf("Invalid Input Year");
exit(0);
}
if(d<=0 || d>ndays)
{
printf("Invalid Input Day");
exit(0);
}
if(m<1 && m>12)
{
printf("Invalid Input Month");
exit(0);
}
if(m==2)
{ if(y%100==0)
```

```
{ if(y%400==0)
ndays=29;
}
else if(y%4==0)
ndays=29;
}
nd=d+1;
nm=m; ny=y;
if(nd>ndays)
{
nd=1;
nm++;
}
if(nm>12)
{
nm=1;
ny++;
}
printf("\n Given date is %d:%d:%d",d,m,y);
printf("\n Next day’s date is %d:%d:%d",nd,nm,ny);
getch( );
}
```

Lab Program: 3

Design, develop, code and run the program in any suitable language to solve the commission problem. Analyze it from the perspective of equivalence class testing, derive different test cases, execute these test cases and discuss the test results.

ALGORITHM

STEP 1: Define lockPrice=45.0, stockPrice=30.0, barrelPrice=25.0

STEP2: Input locks

STEP3: while(locks!=-1) ‘input device uses -1 to indicate end of data goto

STEP 12

STEP4:input (stocks, barrels)

STEP5: compute lockSales, stockSales, barrelSales and sales

STEP6: output(“Total sales:” sales)

STEP7: if (sales > 1800.0) goto STEP 8 else goto STEP 9

STEP8: commission=0.10*1000.0; commission=commission+0.15 * 800.0;

commission = commission + 0.20 * (sales-1800.0)

STEP9: if (sales > 1000.0) goto STEP 10 else goto STEP 11

STEP10: commission=0.10* 1000.0; commission=commission + 0.15 *

(sales-1000.0)

STEP11: Output(“Commission is $”, commission)

STEP12: exit

PROGRAM CODE:

```
#include<stdio.h>

#include<conio.h>

int main()

{

int locks, stocks, barrels, t_sales, flag = 0;

float commission;

clrscr();
```

```
printf("Enter the total number of locks");
scanf("%d",&locks);
if ((locks <= 0) || (locks > 70))
{
flag = 1;
}
printf("Enter the total number of stocks");
scanf("%d",&stocks);
if ((stocks <= 0) || (stocks > 80))
{
flag = 1;
}
printf("Enter the total number of barrelss");
scanf("%d",&barrels);
if ((barrels <= 0) || (barrels > 90))
{
flag = 1;
}
if (flag == 1)
{
printf("invalid input");
getch();
exit(0);
}
t_sales = (locks * 45) + (stocks * 30) + (barrels * 25);
if (t_sales <= 1000)
{
commission = 0.10 * t_sales;
```

```
}
else if (t_sales < 1800)
{
commission = 0.10 * 1000;
commission = commission + (0.15 * (t_sales - 1000));
}
else
{
commission = 0.10 * 1000;
commission = commission + (0.15 * 800);
commission = commission + (0.20 * (t_sales - 1800));
}
printf("The total sales is %d \n The commission is %f",t_sales,
commission);
getch();
return;
}
```

Lab Program: 4

Design, develop, code and run the program in any suitable language to implement the NextDate function. Analyze it from the perspective of equivalence class value testing, derive different test cases, execute these test cases and discuss the test results.

ALGORITHM

STEP 1: Input date in format DD.MM.YYYY

STEP2: if MM is 01, 03, 05,07,08,10 do STEP3 else STEP6

STEP3:if DD < 31 then do STEP4 else if DD=31 do STEP5 else

output(Invalid Date);

STEP4: tomorrowday=DD+1 goto STEP18

STEP5: tomorrowday=1; tomorrowmonth=month + 1 goto STEP18

STEP6: if MM is 04, 06, 09, 11 do STEP7

STEP7: if DD<30 then do STEP4 else if DD=30 do STEP5 else

output(Invalid Date);

STEP8: if MM is 12

STEP9: if DD<31 then STEP4 else STEP10

STEP10: tomorrowday=1, tommorowmonth=1, tommorowyear=YYYY+1;

goto STEP18

STEP11: if MM is 2

STEP12: if DD<28 do STEP4 else do STEP13

STEP13: if DD=28 & YYYY is a leap do STEP14 else STEP15

STEP14: tommorowday=29 goto STEP18

STEP15: tommorowday=1, tomorrowmonth=3, goto STEP18;

STEP16: if DD=29 then do STEP15 else STEP17

STEP17: output("Cannot have feb", DD); STEP19

STEP18: output(tomorrowday, tomorrowmonth, tomorrowyear);

STEP19: exit

PROGRAM CODE:

```
#include<stdio.h>
#include<conio.h>
main( )
{
int month[12]={31,28,31,30,31,30,31,31,30,31,30,31};
int d,m,y,nd,nm,ny,ndays;
clrscr( );
printf("enter the date,month,year");
scanf("%d%d%d",&d,&m,&y);
ndays=month[m-1];
if(y<=1812 && y>2012)
{
printf("Invalid Input Year");
exit(0);
}
if(d<=0 || d>ndays)
{
printf("Invalid Input Day");
exit(0);
}
if(m<1 && m>12)
{
printf("Invalid Input Month");
exit(0);
}
if(m==2)
{
```

```
if(y%100==0)
{ if(y%400==0)
ndays=29;
}
else if(y%4==0)
ndays=29;
}
nd=d+1;
nm=m;
ny=y;
if(nd>ndays)
{
nd=1;
nm++;
}
if(nm>12)
{
nm=1;
ny++;
}
printf("\n Given date is %d:%d:%d",d,m,y);
printf("\n Next day’s date is %d:%d:%d",nd,nm,ny);
getch( );
}
```

Lab Program 5

Design , Develop ,Code & Run the program in any suitable language to implement triangle problem

```
#include <stdio.h>
#include<conio.h>

int main()
{
  int a,b,c,c1,c2,c3;
do {
printf("Enter sides of triangle:\n");
Scanf("%d %d %d",&a,&b,&c);
c1=((a>=1) && (a<=10));
c2=((b>=1) && (b<=10));
c3=((c>=1) && (c<=10));

if(!c1)
printf("Value of a is out of range:");
if(!c2)
printf("Value of b is out of range:");
if(!c3)
printf("Value of b is out of range:");
}
while(!c1||!c2||!c3);
if((a+b)>c && (b+c)>a && (c+a)>b)
{
if(a==b && b==c)
printf("Triangle is Equilateral \n);
else if(a!=b && b!=c && c!=a)
printf("Triangle is Scalar \n);
else
printf("Triangle is Isosceles \n);
}
```

```
else
printf("Triangle Can't br formed \n");
getch();
return 0;
}
```

PART-B

Lab Program 6

Write a script to login a specific web page.

```
package pgm6;

import com.thoughtworks.selenium.*;

public class login extends SeleneseTestCase

{ public void setUp() throws Exception

{ setUp("http://php.thedemosite.co.uk/","*firefox");

}

public void

testloginlogout(){ selenium.setSpeed("2000");

selenium.open("/login.php");

selenium.windowMaximize();

selenium.type("//input[@name='username']", "mytest");

selenium.type("//input[@name='password']", "mytest");

selenium.click("//input[@type='button']");

selenium.waitForPageToLoad("50000");

}
```

Lab Program-7

Write a script to count number of items present on a desktop.

```
Set fso = createobject("Scripting.FileSystemObject")

DesktopPath = CreateObject("WScript.Shell").SpecialFolders("Desktop")

'Files count

msgbox "The number of files on the desktop is : " &fso.GetFolder(DesktopPath).Files.Count

'Folders count

msgbox "The number of folders on the desktop is : "
&fso.GetFolder(DesktopPath).SubFolders.Count

//count.vbs//

package pgm13;

import java.io.*;

public class desktop_item_count

{ public static void main(String[] args)

{ try {

Runtime.getRuntime().exec("wscript  C:/Users/Savitha/Desktop/count.vbs");

} catch (IOException e)

{ System.exit(0);

}

}

}
```

Lab Program 8

Write a script to update 06 student records into table into Excel file.

```
   import  java.io.File;
import java.io.IOException;
import jxl.Cell;
import jxl.CellType;
import jxl.Sheet;
import jxl.Workbook;
import jxl.read.biff.BiffException;


public  class student_excel_read
{
     private String inputFile;
  public  void  setInputFile(String  inputFile)
  {
    this.inputFile=inputFile;
  }
 public void read() throws IOException
 {
    File inputWorkbook=new File(inputFile);
    Workbook w;
    boolean flag=false;
    int count=0;
    try
    {
          w=Workbook.getWorkbook(inputWorkbook);
          Sheet sheet=w.getSheet(0);
          for(int j=0;j<sheet.getRows();j++)
```

```
        {
         for(int i=0;i<sheet.getColumns();i++)
         {
              Cell cell = sheet.getCell(i,j);
               if(cell.getType()==CellType.NUMBER)
               {
                  if(Integer.parseInt(cell.getContents())>60)
                  {
                              flag=true;
                              if(flag==true)
                              {
                                   count++;
                                   flag=false;
                              }
                              break;
                   }
                  }
              }
   }
System.out.println("Total no of students who scored more than 60 in one or more subjects
is:"+count);
        }


   catch(BiffException  e)
   {
         e.printStackTrace();
   }
   }
```

```
public static void main(String[] args)throws IOException
 {
    student_excel_read test=new student_excel_read();
    test.setInputFile("c:/student.xls");
    test.read();
 }
}
```

Lab Program 9

Write a script to provide total number of objects present / available on the page.

Filename: objectcount.java

```
import com.thoughtworks.selenium.*;
@SuppressWarnings("deprecation")
public class objectcount extends SeleneseTestCase
{
public void setUp() throws Exception
{
	setUp("http://php.thedemosite.co.uk/","*firefox");
}

public void testloginlogout()
{
selenium.setSpeed("1000");
selenium.open("http://php.thedemosite.co.uk/");
selenium.waitForPageToLoad("30000");
selenium.windowMaximize();
int num = selenium.getXpathCount("//p").intValue();
System.out.println("The number of option elements present are " +num);
}

public void tearDown() throws Exception
{
	selenium.stop();
}
}
```

Lab Program 10

Write a script to get the number of items in a list / combo box.

```
<html>
<body>
    <select>
        <option>Volvo</option>
        <option>Express</option>
        <option>Mercedes</option>
        <option>RajaHamsa</option>
    </select>
</body>
</html>
```

Write the below code in eclipse and before running this code makes sure your selenium RC server is running.

Filename: combo.java

```
import com.thoughtworks.selenium.*;
public class combo extends SeleneseTestCase
{
public void setUp() throws Exception
{
        selenium = new DefaultSelenium("localhost", 4444, "*chrome","file:///C:/combo.html");
        selenium.start();
}
public void testloginlogout()
{
        selenium.setSpeed("1000");
        selenium.open("file:///C:/combo.html");
        selenium.waitForPageToLoad("30000");
        selenium.windowMaximize();
        String[] selectelements = new String[1000];
        selectelements= selenium.getSelectOptions("//select");
System.out.println("The number of options present in the given select box is “+selectelements.length);
}

public void tearDown() throws Exception
{
        selenium.stop();
}
}
```